



Intro to AI

part2

Recap

what we have seen earlier



- Introduction (What is AI, history, The value of AI, and ANI vs AGI).
- ML (Definitions, types of ML, what makes AI so powerful recently?).
- Supervised (Examples, how most of AI problems is around it).
- Un-supervised (why it's needed, and problem it solves in the real world).
- Data (how it worth?, types of data, its resources, Use and misuse of data, data is messy).
- Data Science

Agenda



1. Deep Learning (intro, history, why it's needed over ML, and its applications).
2. AI and related disciplines.
3. RL (Intro, how it's important to achieve AGI, examples).
4. Computer vision.
5. NLP (intro, it's examples in the real world).
6. What ML can and can't do? (examples, strengths, and weaknesses of ML).
7. What makes you a good data scientist?

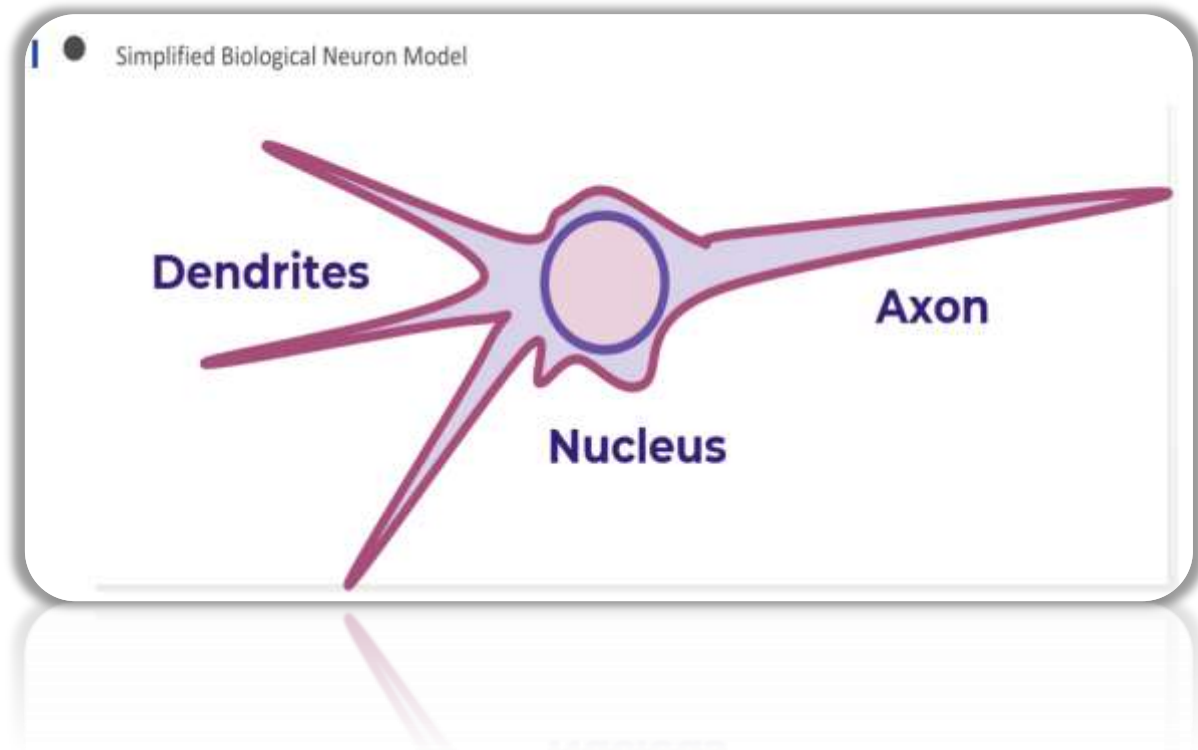
Intro to Deep Learning (DL)

- At a very basic level, deep learning is a **machine learning** technique that teaches a computer to filter inputs.
- Deep learning is inspired by the way that the **human brain** filters information!
- Essentially, deep learning is a part of the machine learning family that's based on **learning data representations** (rather than task-specific algorithms).
- DL closely related to a class of theories about brain development proposed by **cognitive neuroscientists** in the early '90s.
- In the human brain, there are about 100 billion neurons, and each neuron is connected to about 100,000 of its neighbors. Essentially, that is what we're trying to create, but in a way and *at a level that works for machines*.

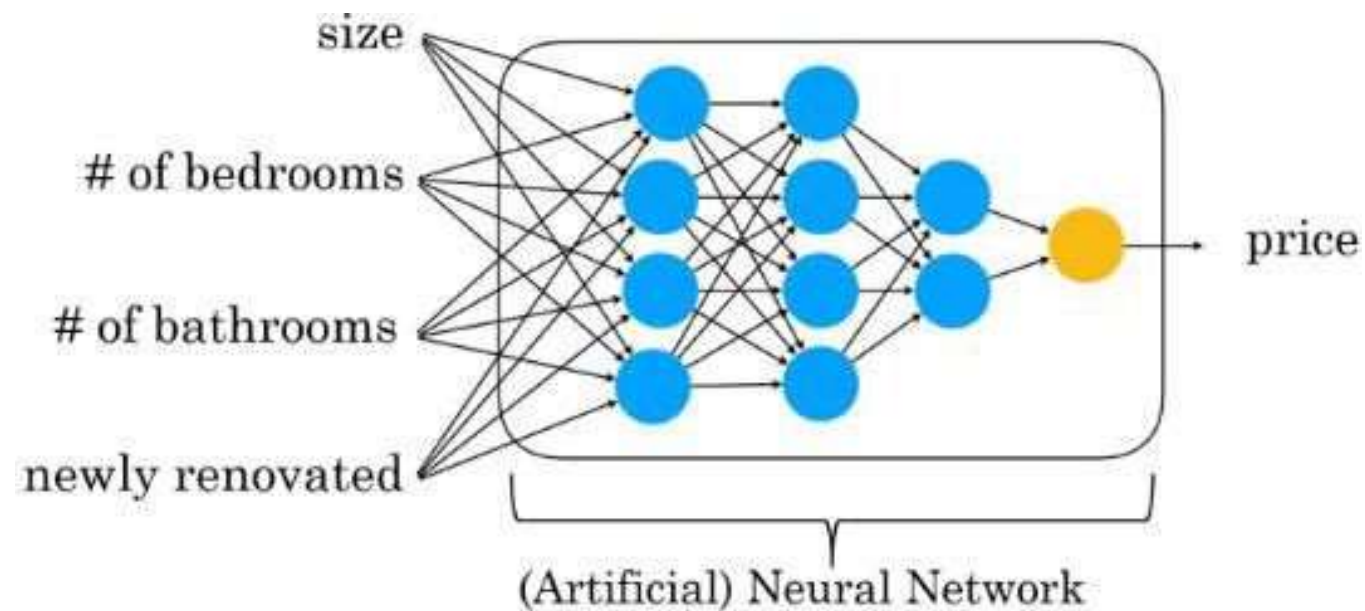


Artificial neural networks (ANN).

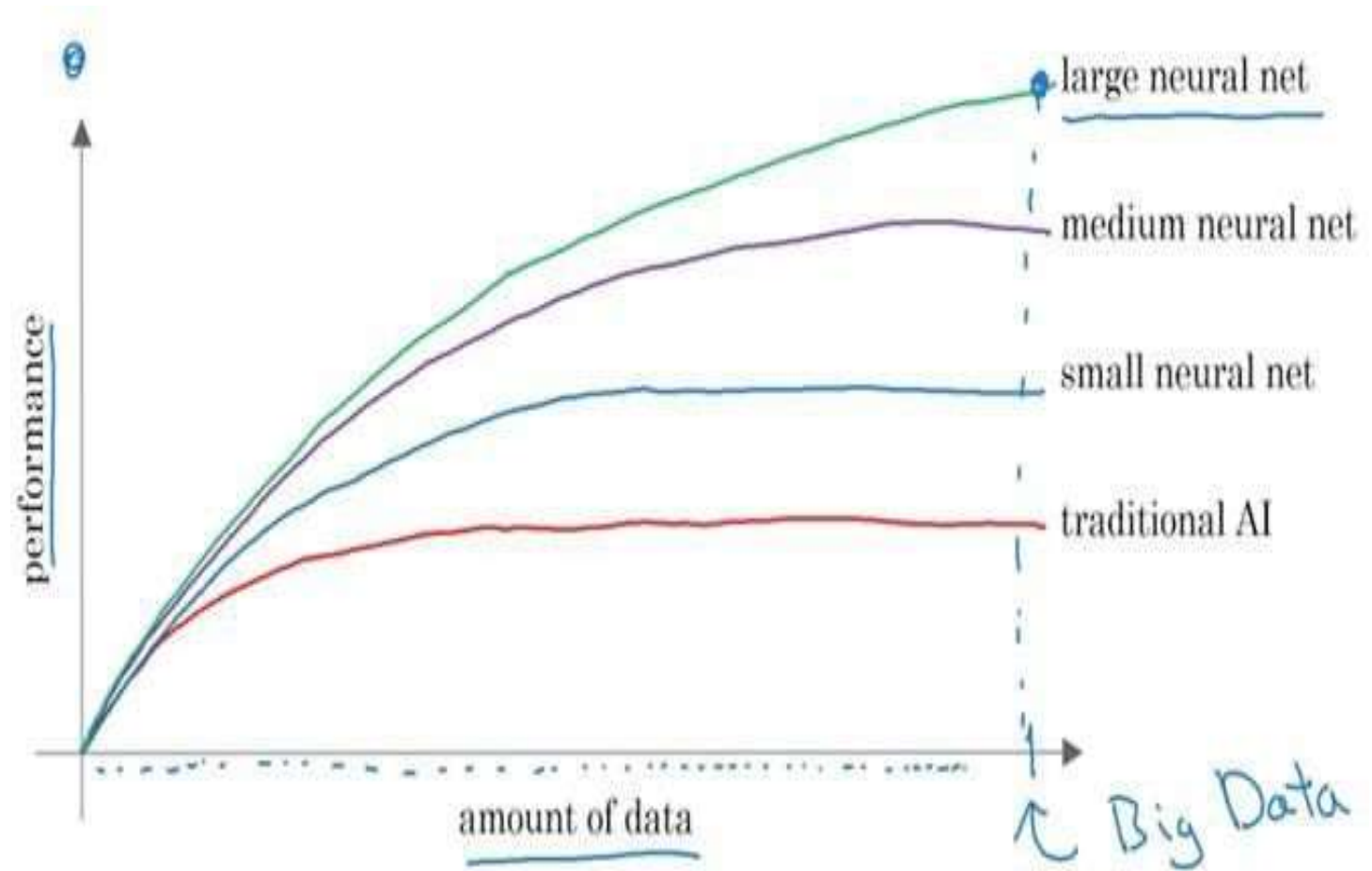
- As we said before DL is just a way to **mimic** our human brains.
- And this **pic** is a simplified version of biological neuron and based on it we build or neural networks models.
- When we use more than one or two neurons we named it as a **deep neural networks**, thus the name DL cam from.



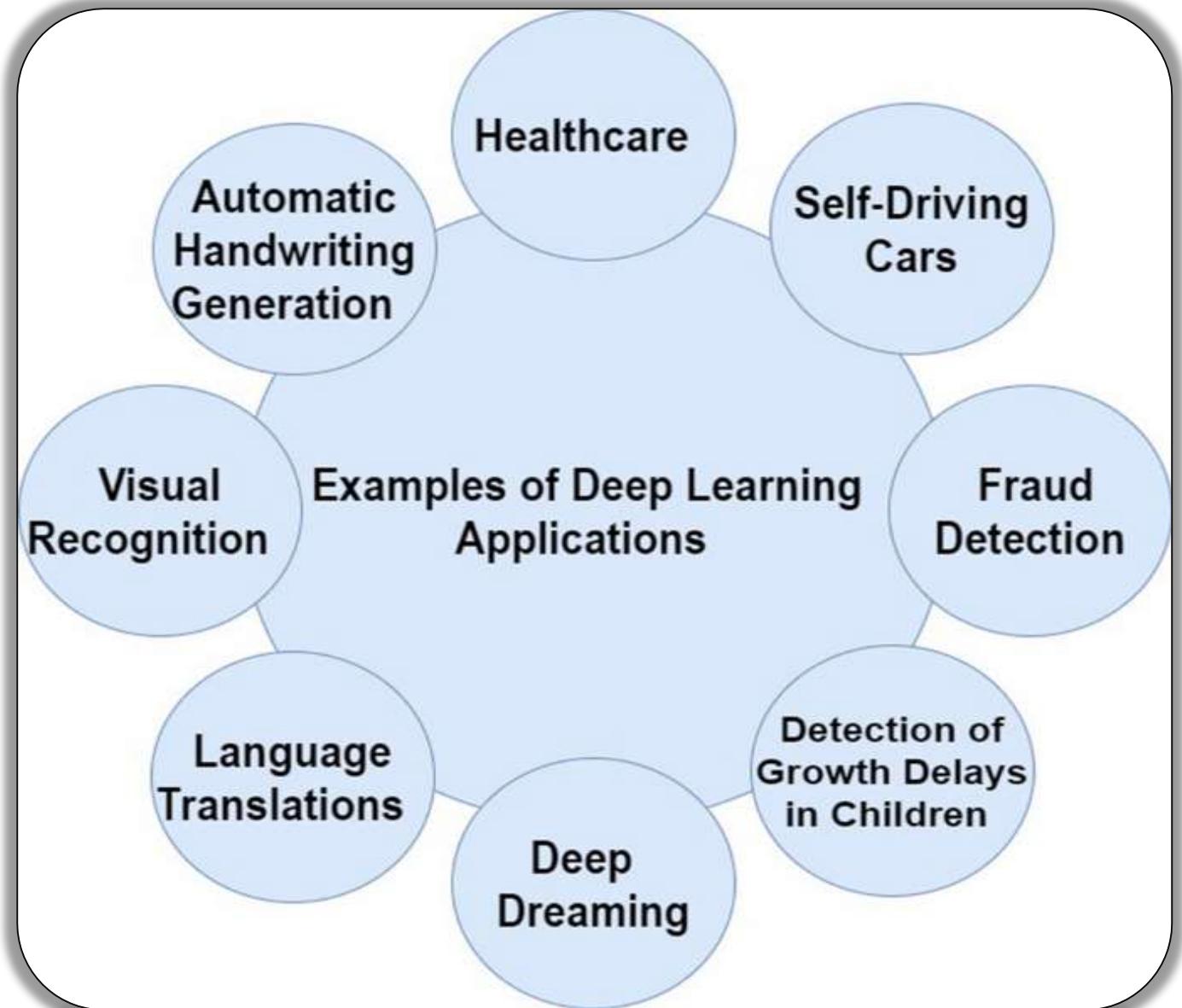
Deep Learning Example



Why it's needed over ML



DL Applications



AI And Related Disciplines let's draw it.





Reinforcement Learning (RL)

- How do we pet our dogs/cats?, What do you do to give it a **reward** or **punish** it?
- Reinforcement learning is a machine learning training method based on **rewarding desired** behaviors and/or **punishing undesired** ones.
- In general, a reinforcement learning is an **agent** able to *perceive* and interpret its environment, take actions and learn through trial and error.
- Developers devise a method of rewarding desired behaviors and punishing negative behaviors (Maze Example).
- So, in short RL agents' goals are to **maximize** the reward it achieves.

Let's watch this video 😊

Main ML Types, And Differences

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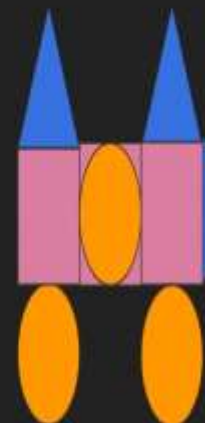
Types of Learning

- Unsupervised learning



Types of Learning

- Reinforcement learning (RL)



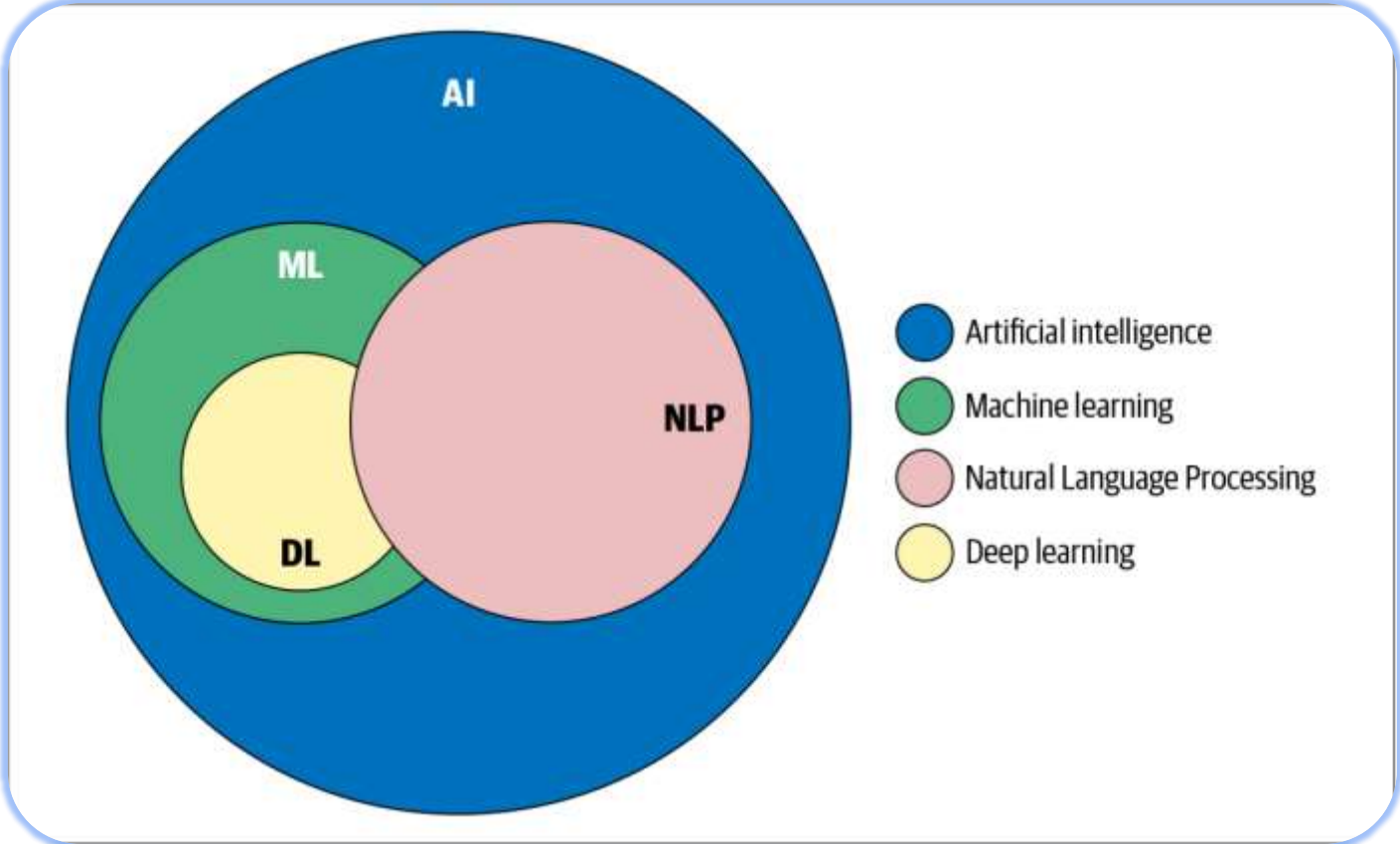
Computer Vision (CV)

- Computer vision defines the field that enables devices to acquire, process, understand, and analyze **digital images and videos** and extract useful information.
- CV is as **sub-field** of deep learning, and most of the models we will use in this part as we will see is deep neural networks called **CNN**.
- CNN just **mimic** the way humans recognize things or our visual cortex mechanism.
- Main **tasks**:

Segmentation, Classification, and Detection

Segmentation	Classification	Detection
Good at delineating objects	Is it a cat or a dog?	Where does it exist in space?
Used in self-driving vehicles	Classifies with precision	Recognizes things for safety

- How machines process and understand human **language**?
- In theory, we can understand and even predict human behaviors using that info we get from **text**.
- So, NLP is a field of Artificial Intelligence that gives the machines the ability to read, understand and derive meaning from human languages.
- And for you **as a data scientist** you will need NLP more than CV most of the time.



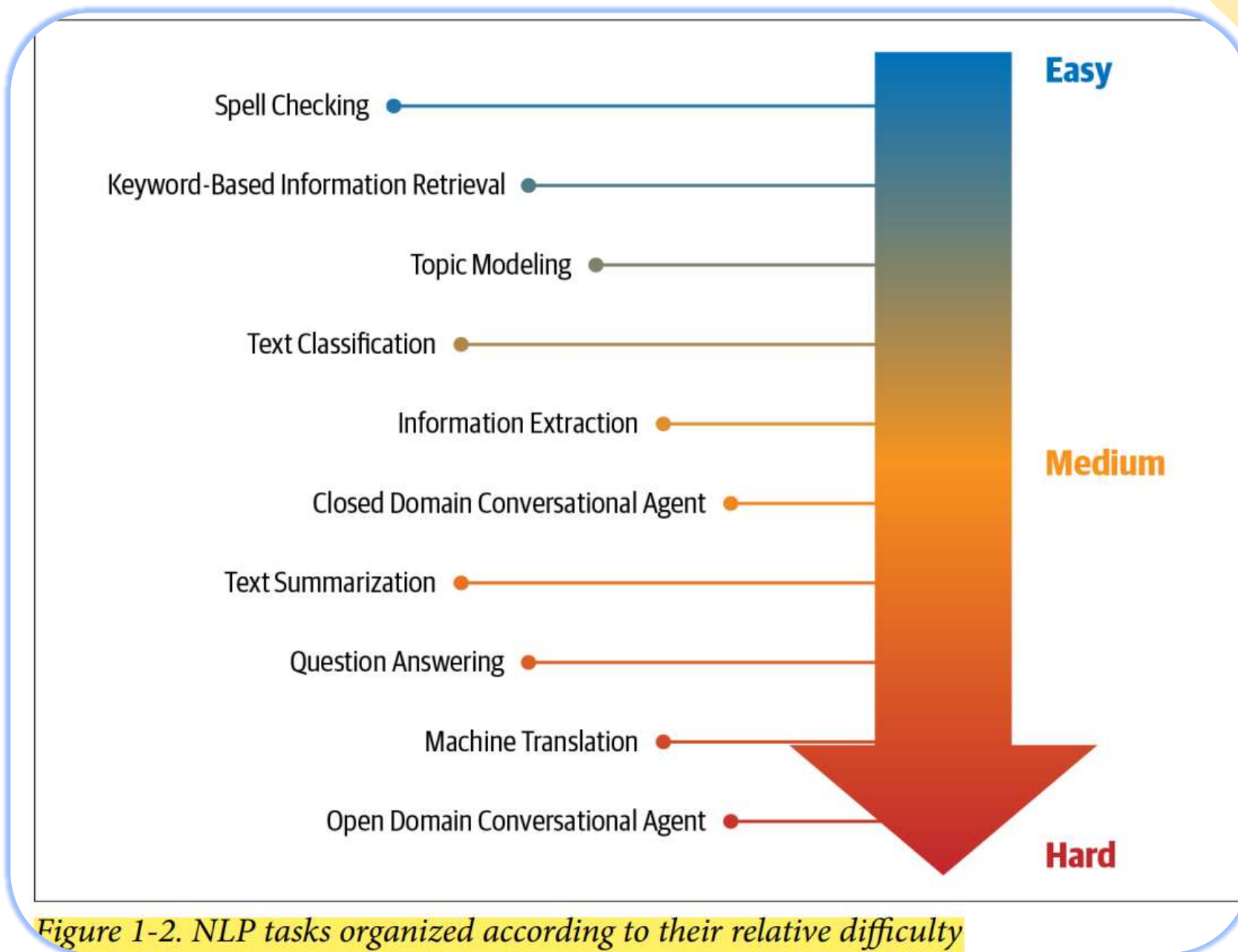


Figure 1-2. NLP tasks organized according to their relative difficulty

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Why NLP Is Challenging?

- What makes NLP a challenging problem domain? **The ambiguity and creativity of human language.**
- Ambiguity: means **uncertainty of meaning**. Most human languages are inherently ambiguous. Consider the following sentence: “I made her duck.”. *The first one* is: I cooked a duck for her. *The second meaning* is: I made her bend down to avoid an object 😊.
- Creativity: Poems are a great example of creativity in language. Making machines understand creativity is a hard problem not just in NLP, but in AI in general.
- So, NLP is really **fluctuating** field and scientist really infested about it, and for you as DS you will find how its important for you in practice.

Them: "AI is going to take over the world and kill us"

Meanwhile AI:



ML Drawbacks

What machine learning today can and cannot do

Customer Services Model

The toy arrived two days late, so I wasn't able to give it to my niece for her birthday.
Can I return it?



"Refund request"



Oh, sorry to hear that.
I hope your niece had a good birthday.
Yes, we can help with....

Input text → Refund/Shipping/Other

A → B

Making machines understand creativity is a hard problem in AI in general

Examples of what ML can and can't do?



Identifying the intent of the customer - **Possible**



Writing an emphatic response to customer's email – **Not possible or difficult**

X-ray diagnosis

1

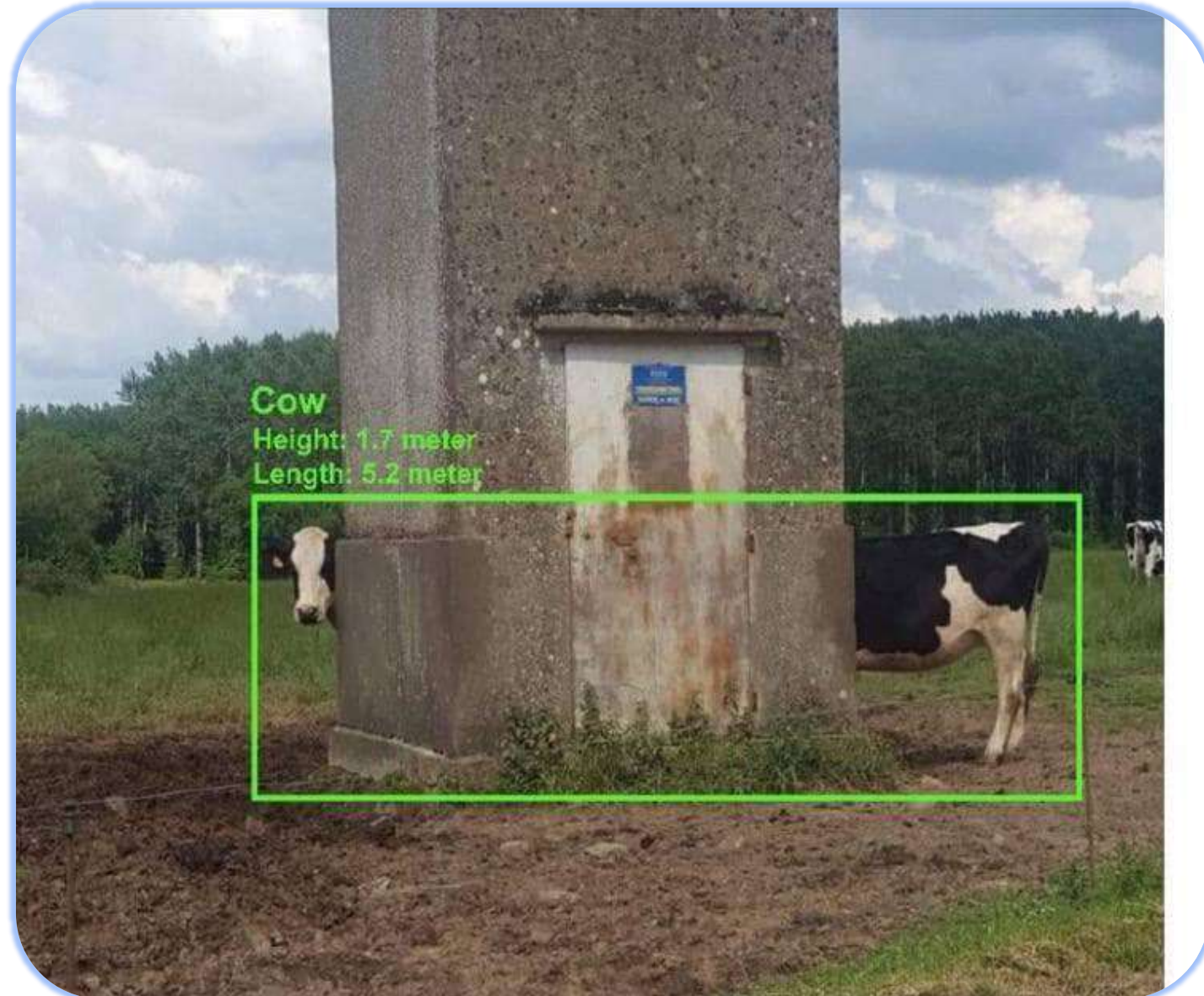
Diagnosing a disease from X-ray images— **possible**

2

Diagnosing a disease after reading a book- **difficult or almost impossible.**

Strengths and weakness of ML

- **Works when,**
 - Learning a simple concept.
 - Lots of data available.
- **Doesn't work when,**
 - Learning a complex concept (as we see in the last examples).
 - Asked to work on new type of data such as X-ray images in different conditions and angles.



**What makes you a good
data scientist?**

- Two types of skills you need:
 1. Tools Skills.
 2. Core Skills.
- For **tools (technical) skills**: here is the practical part come in, you need to learn the **main tools** you will use like:
 1. Python/R as a **programming language** and its DS modules e.g. (numpy, pandas and matplotlib) in py and how to use them in DS projects flow.
 2. **ML** or/and NLP.
 3. SQL: to be able to **query your data** from database for instance.
 4. **Excel** (Foundations and main formulas).
 5. Power BI/Tableau for **data VIZs**.
 6. Math (Mainly **probability and statistics**).
- This is the main tools/technical skills will be required from you as a junior data scientist at most.

The Core Skills Is Really Make A Unique DS.

- Mastering this tools in the previous slides is important, but to have this core skills is the beautiful part to be a data scientist.
- So, what is this skills about?
 - **Business Understanding**
 - **Analytical Thinking**
 - **Communication**
- We can see about it, it's the way or the **mindset** you have in work.

DONE



ANY QUESTIONS?